

Coarse Classes, Hyperdegrees, and the Cone Filter

Two results at the junction of the coarse degrees with the hyperarithmetical hierarchy and with Martin's cone theorem

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Summary

Section 12 of the research plan relates the Turing degrees to large countable ordinals, through the hyperarithmetical hierarchy, and to large cardinals, through Martin's cone theorem, and it lists directions (O1)–(O5) for the coarse track. This report settles two of them.

(O1), negatively. There are coarse-equivalence classes with no representative of least *hyperdegree* (Theorem 1.1, Corollary 1.2). More precisely, for every oracle Z and every disagreement budget there are sets A, B , each Cohen generic over $L_{\omega_1^Z}[Z]$, with $A \triangle B$ obeying the budget, such that $\Delta_1^1(Z \oplus A) \cap \Delta_1^1(Z \oplus B) = \Delta_1^1(Z)$. Hence every hyperdegree is the unattained infimum of the hyperdegree spectrum of a coarse class (Corollary 1.3). This is not a relabelling of the Turing result: the dyadic codes, which refute C1 for Turing degrees, *do* have least hyperdegrees, and the cone-avoiding compactness theorem, on which the literature route to C1 rests, is *false* for $\leq_h$ (Theorem 1.4). The proof replaces finite use by the forcing relation in the one-bit bridge lemma of the synthesis.

(O4), negatively. Martin's cone theorem has no analogue for the coarse degrees. The set of functions whose coarse class has a least Turing degree is Π_1^1 , invariant under both coarse equivalences, and it meets every coarse cone, as does its complement (Theorem 1.5). This is a theorem of ZF; no determinacy hypothesis can repair it.

All proofs are conventional and have not been refereed or formalized. Novelty has not been established. The facts about Cohen forcing over $L_{\omega_1^{\text{CK}}}$ that the proof of Theorem 1.1 uses are quoted from the literature, from memory; they are isolated in Section 2.2 so that they can be checked.

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1 Results

Notation follows the synthesis [1]: $f \approx g$ means that f, g agree off a set of density zero, $\text{CD}(f)$ is the class of coarse descriptions of f , $\leq_{\text{nc}}$ and $\leq_{\text{uc}}$ are nonuniform and uniform coarse reducibility, $J(g)$ is the block code, and $\mathcal{R}(A)(n) = A(v_2(n+1))$ is the dyadic code. $X \leq_h Y$ means that X is hyperarithmetical in Y , that is $X \in \Delta_1^1(Y)$; $\deg_h$ denotes hyperdegree; ω_1^Y is the least ordinal with no Y -computable copy, and $\mathcal{O}^Y$ is the hyperjump of Y . For a total f put

$$\text{Spec}_r^h(f) = \{\deg_h(g) : g \equiv_r f\} \quad (r \in \{\text{uc}, \text{nc}\}), \quad \text{Core}_h(f) = \{A \subseteq \mathbb{N} : (\forall D \in \text{CD}(f)) A \leq_h D\}.$$

A *budget* is a family $\mathfrak{B}$ of finite sets as in [1, Definition 8.1]; the definition is recalled in Section 4. The reader may keep in mind the counting budget $|E \cap [0, k]| \leq \lfloor \log_2(k+1) \rfloor$.

Theorem 1.1 (Hyperdegree minimal pairs in one coarse class). Let $Z \subseteq \mathbb{N}$ and let $\mathfrak{B}$ be a budget. There are sets A, B such that

- (i) A and B are each Cohen generic over $L_{\omega_1^Z}[Z]$ in the sense of Section 2.2; in particular $\omega_1^{Z \oplus A} = \omega_1^{Z \oplus B} = \omega_1^Z$, and neither A nor B has a coarse description in $\Delta_1^1(Z)$;
- (ii) $A \triangle B$ obeys $\mathfrak{B}$; in particular $A \approx B$ when $\mathfrak{B}$ forces density zero;
- (iii) $\Delta_1^1(Z \oplus A) \cap \Delta_1^1(Z \oplus B) = \Delta_1^1(Z)$.

Consequently $\deg_h(Z \oplus A)$ and $\deg_h(Z \oplus B)$ are incomparable, lie strictly above $\deg_h(Z)$, and have greatest lower bound $\deg_h(Z)$.

Corollary 1.2 (No least hyperdegree). With $Z = \emptyset$ and a budget forcing density zero, the uniform and the nonuniform coarse class of A contain no representative of least hyperdegree, although the two witnesses satisfy $\omega_1^A = \omega_1^B = \omega_1^{\text{CK}}$. A fortiori they contain no representative of least Turing degree.

Corollary 1.3 (Every hyperdegree is an unattained infimum). For every Z there is a binary F such that for $r \in \{\text{uc}, \text{nc}\}$ the infimum of $\text{Spec}_r^h(F)$ in the hyperdegrees exists and equals $\deg_h(Z)$, but $\deg_h(Z) \notin \text{Spec}_r^h(F)$; moreover $\text{Core}_h(F) = \Delta_1^1(Z)$.

Theorem 1.4 (Dyadic codes; failure of hyperarithmetical compactness). For every set A :

- (a) $\text{Core}_h(\mathcal{R}(A)) = \Delta_1^1(A)$, and $\text{Spec}_{\text{uc}}^h(\mathcal{R}(A)) = \text{Spec}_{\text{nc}}^h(\mathcal{R}(A)) = \{c : c \geq \deg_h(A)\}$. So the coarse classes of $\mathcal{R}(A)$ have least hyperdegree $\deg_h(A)$, whereas they have a least Turing degree only when $A \leq_T \emptyset'$.
- (b) If A is not hyperarithmetical, then $A \in \text{Core}_h(\mathcal{R}(A))$ although for every $\varepsilon > 0$ there is a computable Y with $\delta(\mathcal{R}(A), Y) < \varepsilon$, and of course $A \not\leq_h Y$. Thus the robust-radius characterization of the core [1, Theorem 4.8], equivalently the cone-avoiding compactness theorem [2, Theorem 3.7], fails when $\leq_T$ is replaced by $\leq_h$; so does its corollary that $\gamma(X) = 1$ implies a trivial core [2, Theorem 4.3].
- (c) The ordinal $o(f) = \min\{\omega_1^g : g \equiv_{\text{nc}} f\}$ satisfies $o(\mathcal{R}(A)) = \omega_1^A$. In particular every representative g of the class of $\mathcal{R}(\mathcal{O})$ has $\omega_1^g > \omega_1^{\text{CK}}$, although by [1, Theorem 6.3] a representative need not compute $\mathcal{O}$: it suffices that its jump does. Every countable admissible ordinal greater than ω is $o(f)$ for some f .

Theorem 1.5 (No cone theorem for coarse degrees). Let $\mathcal{A} = \{f \in \mathbb{N}^{\mathbb{N}} : \text{Spec}_{\text{nc}}(f) \text{ has a least element}\}$. Then:

- (a) $\mathcal{A}$ is a Π_1^1 set of reals, closed under $\equiv_{\text{nc}}$ and under $\equiv_{\text{uc}}$;
- (b) for every $h \in \mathbb{N}^{\mathbb{N}}$ there are $f_0 \in \mathcal{A}$ and $f_1 \notin \mathcal{A}$ with $h \leq_{\text{uc}} f_0$ and $h \leq_{\text{uc}} f_1$, hence also $h \leq_{\text{nc}} f_0, f_1$.

Consequently, in ZF, neither $\mathcal{A}$ nor its complement contains a cone of uniform or of nonuniform coarse degrees: the cone filter on either structure is not an ultrafilter, not even on Π_1^1 sets. Under AD the push-forward of the Martin measure along $x \mapsto [J(x)]_{\text{nc}}$ is a countably complete ultrafilter on the nonuniform coarse degrees that properly extends the cone filter and concentrates on $\mathcal{A}$.

How the results bear on the plan. Theorem 1.1 answers (O1) and shows that the phenomenon behind C1 persists at the level where ω_1^{CK} is the natural bound: the witnesses do not raise ω_1 . Theorem 1.4 is the calibration recorded as (O2), and it explains why (O1) is not a consequence of the Turing case: compactness is a finite-use phenomenon. Theorem 1.5 answers (O4) and identifies exactly where Martin’s strategy-stealing argument breaks (Remark 5.1). Section 6 records what remains, including an ordinal-indexed question about which jump-cones are coarse spectra.

Related work. Igusa [5] characterizes the hyperarithmetic reals through the generic degrees of density-one sets and shows that generic reducibility is Π_1^1 -complete; that paper concerns generic rather than coarse reducibility and does not consider hyperdegrees of equivalence classes. We have not found the statements above in the literature, but no systematic search was made.

2 Background

2.1 Coarse notions

We use from [1]: density-zero changes preserve the class of coarse descriptions and are uniform equivalences (Lemma 2.1); a representative $g \equiv_{\text{nc}} f$ computes a member of $\text{CD}(f)$ (Lemma 2.3, self-application); every $D \in \text{CD}(J(g))$ computes g , and $g \leq_{\text{T}} h$ iff $J(g) \leq_{\text{nc}} J(h)$ (Lemma 3.4, Proposition 3.5); the class of f has a least Turing degree iff some set A computes a member of $\text{CD}(f)$ and is computable from every member of $\text{CD}(f)$ (Theorem 3.6(iv)); B computes a coarse description of $\mathcal{R}(A)$ iff $A \leq_{\text{T}} B'$ (Theorem 6.3); and for every Z there is a binary $F = J(Z) \oplus W$ whose coarse classes have no least Turing degree (Theorem 9.2).

2.2 Hyperarithmetic facts

The following are standard; see [3], Chapters II–IV, and [4]. They are quoted from memory and should be checked against those sources. All of them relativize to an arbitrary real Z , with $\omega_1^{\text{CK}}, \mathcal{O}, L_{\omega_1^{\text{CK}}}$ replaced by $\omega_1^Z, \mathcal{O}^Z, L_{\omega_1^Z}[Z]$ and “hyperarithmetic” by “hyperarithmetic in Z ”.

- (H1) $X \leq_{\text{T}} Y'$ implies $X \leq_h Y$; $\leq_{\text{T}}$ implies $\leq_h$; ω_1^X depends only on $\deg_h(X)$ and is monotone in it. *Spector’s criterion:* $\omega_1^X > \omega_1^{\text{CK}}$ iff $\mathcal{O} \leq_h X$. *Sacks:* every countable admissible ordinal greater than ω is ω_1^X for some X .

Cohen forcing over $L_{\omega_1^{\text{CK}}}$. Conditions are finite binary strings. The ramified language $\mathcal{L}(\omega_1^{\text{CK}}, \mathcal{G})$ has a set constant $\mathcal{G}$, number variables, ranked set variables x^β for $\beta < \omega_1^{\text{CK}}$ and unranked set variables; a formula is *ranked* if all its set variables are ranked. For a real G the structure $\mathcal{M}(G)$ interprets x^β as ranging over the sets first-order definable over the structure built below β , as in $L_{\omega_1^{\text{CK}}}[G]$. The (strong) forcing relation $p \Vdash \varphi$ is defined as usual, with $p \Vdash \neg \varphi$ iff no $q \supseteq p$ forces φ . A real G is *generic* (over $L_{\omega_1^{\text{CK}}}$) if every sentence of the language is decided by some initial segment of G .

- (H2) *Basic forcing facts.* No condition forces both φ and $\neg\varphi$; forcing is preserved under extension; for every φ the conditions deciding φ are dense. If G is generic then $\mathcal{M}(G) \models \varphi$ iff some $p \subset G$ forces φ . For countably many dense sets of conditions there is a generic G meeting them all, extending any prescribed condition.
- (H3) *Definability.* For each ranked formula $\psi(x)$ with one free number variable, the relation $\{(p, n) : p \Vdash \psi(\bar{n})\}$ is Δ_1^1 .
- (H4) *Preservation and naming.* If G is generic then $\omega_1^G = \omega_1^{\text{CK}}$, and a set X is hyperarithmetical in G iff $X = \{n : \mathcal{M}(G) \models \psi(\bar{n})\}$ for some ranked formula $\psi(x)$.
- (H5) *Hyperarithmetical dense sets are met.* If G is generic and W is a dense hyperarithmetical set of strings, then some initial segment of G lies in W .

3 Hyperdegree spectra of coarse classes

Lemma 3.1 (Spectrum identity for hyperdegrees). For every total f ,

$$\text{Spec}_{\text{uc}}^h(f) = \text{Spec}_{\text{nc}}^h(f) = \{\mathbf{c} : (\exists D \in \text{CD}(f)) \deg_h(D) \leq \mathbf{c}\},$$

and every member is the hyperdegree of a member of $\text{CD}(f)$. Hence $\text{Spec}_r^h(f)$ has a least element iff some $D_0 \in \text{CD}(f)$ satisfies $D_0 \leq_h D$ for every $D \in \text{CD}(f)$.

Proof. If $g \equiv_{\text{nc}} f$ then g computes some $D \in \text{CD}(f)$, so $\deg_h(D) \leq \deg_h(g)$. Conversely let $D \in \text{CD}(f)$ and $D \leq_h C$. Replace C by a set of the hyperdegree of C that computes D in the Turing sense, for instance $C \oplus \text{graph}(D)$, and code it into D on the powers of two: $Y(2^n) = C(n)$ and $Y(k) = D(k)$ otherwise. Then $Y \approx D$, so $Y \in \text{CD}(f)$ and $Y \equiv_{\text{uc}} f$, and $Y \equiv_T C$. For the last statement, a least element is the hyperdegree of some description D_0 by what was just proved, and every description is a representative. $\square$

Lemma 3.2 (Two-witness obstruction for hyperdegrees). Let Z be a set. Suppose f has no coarse description in $\Delta_1^1(Z)$, every member of $\text{CD}(f)$ computes Z , and $D, E \in \text{CD}(f)$ satisfy $\Delta_1^1(D) \cap \Delta_1^1(E) = \Delta_1^1(Z)$. Then for $r \in \{\text{uc}, \text{nc}\}$ the hyperdegree $\deg_h(Z)$ is the greatest lower bound of $\text{Spec}_r^h(f)$ and does not belong to it; in particular there is no least element. Moreover $\text{Core}_h(f) = \Delta_1^1(Z)$.

Proof. Every representative computes a description and hence Z , so $\deg_h(Z)$ is a lower bound. A representative $g \leq_h Z$ would compute a description $D' \leq_T g$, which would lie in $\Delta_1^1(Z)$. A lower bound $\mathbf{c}$ of the spectrum is below $\deg_h(D)$ and $\deg_h(E)$, so a set of degree $\mathbf{c}$ lies in $\Delta_1^1(D) \cap \Delta_1^1(E) = \Delta_1^1(Z)$. Finally $\Delta_1^1(Z) \subseteq \text{Core}_h(f) \subseteq \Delta_1^1(D) \cap \Delta_1^1(E)$. $\square$

Proof of Theorem 1.4. (a) Let B be a coarse description of $\mathcal{R}(A)$, binary after normalization. By [1, Theorem 6.3], $A \leq_T B'$, so $A \leq_h B$ by (H1). Thus $\Delta_1^1(A) \subseteq \text{Core}_h(\mathcal{R}(A))$. The reverse inclusion holds because $\mathcal{R}(A)$ is a description of itself and $\mathcal{R}(A) \equiv_T A$. For the spectrum, every representative computes a description, hence is $\geq_h A$, and $\mathcal{R}(A)$ has hyperdegree $\deg_h(A)$; upward closure is Lemma 3.1. The statement about Turing degrees is [1, Theorem 7.4].

(b) The computable sets Q_K that agree with $\mathcal{R}(A)$ on the columns below K and vanish elsewhere satisfy $\delta(\mathcal{R}(A), Q_K) \leq 2^{-K}$ [1, Proposition 6.5]. If the robust-radius characterization held for $\leq_h$, then $A \in \text{Core}_h(\mathcal{R}(A))$ would give $\eta > 0$ such that $A \leq_h Y$ whenever $\delta(\mathcal{R}(A), Y) < \eta$, and $Y = Q_K$ with $2^{-K} < \eta$ would make A hyperarithmetical. The same example refutes “ $\gamma(X) = 1$ implies $\text{Core}_h(X)$ is the class of hyperarithmetical sets”.

(c) Every representative g satisfies $A \leq_h g$ by (a), so $\omega_1^g \geq \omega_1^A$ by (H1), and $\mathcal{R}(A) \equiv_T A$ attains the bound. For $A = \mathcal{O}$ use Spector's criterion. Given a countable admissible $\alpha > \omega$, take X with $\omega_1^X = \alpha$ by (H1); then $o(\mathcal{R}(X)) = \alpha$. $\square$

Here $\mathbf{c} \geq_h A$ abbreviates $A \leq_h C$ for $C \in \mathbf{c}$.

Remark 3.3 (Why compactness fails). The proof of the robust-radius theorem decodes A from a nearby oracle Y by searching for a finite string τ close to Y on which a Turing functional converges; correctness comes from patching τ onto a genuine description, which preserves the computation because it has finite use [1, Lemma 4.3]. A hyperarithmetic reduction has no finite use: majority decoding of a column of $\mathcal{R}(A)$ is a limit, and a limit is destroyed by changing the oracle on a set of small positive density but survives changes on a set of density zero. Theorem 1.4(b) says that this is not an artefact of the proof.

4 Proof of Theorem 1.1

We give the proof for $Z = \emptyset$; the general case is the same argument with every notion relativized to Z , as provided for in Section 2.2.

4.1 The forcing

A *budget* is a decidable family $\mathfrak{B}$ of finite subsets of $\mathbb{N}$, containing $\emptyset$ and closed under subsets, with a computable function m such that for $E \in \mathfrak{B}$ and $L \in \mathbb{N}$: $m(E, L) \geq L$, $m(E, L) > \max E$, and $E \cup \{k\} \in \mathfrak{B}$ for all $k \geq m(E, L)$. A set V *obeys* $\mathfrak{B}$ if $V \cap [0, n) \in \mathfrak{B}$ for all n . For equal-length strings σ, τ let $\Delta(\sigma, \tau) = \{i : \sigma(i) \neq \tau(i)\}$.

Let $\mathbb{P}$ be the set of pairs $p = (\sigma, \tau)$ of binary strings of equal length with $\Delta(\sigma, \tau) \in \mathfrak{B}$, ordered by coordinatewise extension. Call $p = (\sigma, \tau)$ *padded* if $\Delta(\sigma, \tau) \cup \{k\} \in \mathfrak{B}$ for every $k \geq |\sigma|$. Write $\pi_0(\sigma, \tau) = \sigma$ and $\pi_1(\sigma, \tau) = \tau$.

Lemma 4.1. (a) (Projection.) If $(\sigma, \tau) \in \mathbb{P}$ and $\alpha \supseteq \sigma$, then $(\alpha, \tau \frown u) \in \mathbb{P}$ extends (σ, τ) , where $\alpha = \sigma \frown u$; symmetrically for the second coordinate. Hence if W is a dense set of strings then $\pi_i^{-1}[W]$ is dense in $\mathbb{P}$.

(b) (Padding.) Every condition has a padded extension, obtained by appending zeros to both coordinates.

(c) (One-bit reserve.) If (σ, τ) is padded and u, v are strings of equal length that differ in at most one position, then $(\sigma \frown u, \tau \frown v) \in \mathbb{P}$, and it is padded if $u = v$.

Proof. (a) The disagreement set is unchanged. (b) Append zeros up to length $m(\Delta(\sigma, \tau), |\sigma|)$. (c) The disagreement set is $\Delta(\sigma, \tau)$ or $\Delta(\sigma, \tau) \cup \{k\}$ with $k \geq |\sigma|$. $\square$

4.2 The bridge lemma in forcing form

Fix ranked formulas $\psi^0(x), \psi^1(x)$ and write φ_n^i for the sentence $\psi^i(\bar{n})$. All forcing below is Cohen forcing on single strings. A condition $(\alpha, \beta) \in \mathbb{P}$ is a *cross-disagreement* (for ψ^0, ψ^1) if for some n either $\alpha \Vdash \varphi_n^0$ and $\beta \Vdash \neg \varphi_n^1$, or $\alpha \Vdash \neg \varphi_n^0$ and $\beta \Vdash \varphi_n^1$.

Lemma 4.2 (Bridge lemma). Let $p^* = (\sigma^*, \tau^*) \in \mathbb{P}$ be padded and suppose that no extension of p^* in $\mathbb{P}$ is a cross-disagreement. Then for every n , all strings $\alpha \supseteq \sigma^*$ that decide φ_n^0 decide it the same way.

Consequently the set

$$F = \{n : (\exists \alpha \supseteq \sigma^*) \alpha \Vdash \varphi_n^0\} = \{n : \neg(\exists \alpha \supseteq \sigma^*) \alpha \Vdash \neg \varphi_n^0\}$$

is hyperarithmetical.

Proof. Fix n and let $\alpha_0 = \sigma^* \frown u_0$ and $\alpha_1 = \sigma^* \frown u_1$ decide φ_n^0 . Extending the shorter one, which preserves what it forces by (H2), we may assume $|u_0| = |u_1|$. Choose a path $u_0 = w_0, w_1, \dots, w_t = u_1$ of strings of that length in which consecutive strings differ in exactly one position.

One suffix serves the whole path. We find a string ρ such that every $\sigma^* \frown w_i \frown \rho$ decides φ_n^0 and every $\tau^* \frown w_i \frown \rho$ decides φ_n^1 . Treat these $2(t+1)$ requirements one at a time: if the current common suffix is ρ_0 , the deciding conditions are dense (H2), so some v makes the next string, extended by $\rho_0 \frown v$, decide its sentence; replace ρ_0 by $\rho_0 \frown v$ everywhere. Decisions already obtained persist under extension.

Let $a_i \in \{0, 1\}$ record whether $\sigma^* \frown w_i \frown \rho$ forces φ_n^0 or its negation, and b_i likewise for $\tau^* \frown w_i \frown \rho$ and φ_n^1 . By Lemma 4.1(c) the pairs

$$(\sigma^* \frown w_i \frown \rho, \tau^* \frown w_i \frown \rho) \quad \text{and} \quad (\sigma^* \frown w_i \frown \rho, \tau^* \frown w_{i+1} \frown \rho)$$

are conditions extending p^* , since $w_i \frown \rho$ and $w_{i+1} \frown \rho$ differ in one position. Neither is a cross-disagreement, so $a_i = b_i$ and $a_i = b_{i+1}$. Hence $a_0 = a_1 = \dots = a_t$. Since $\alpha_0 \subseteq \sigma^* \frown w_0 \frown \rho$ and $\alpha_1 \subseteq \sigma^* \frown w_t \frown \rho$, and forcing persists while no condition forces a sentence and its negation, α_0 and α_1 decide φ_n^0 the same way.

The two descriptions of F agree because some $\alpha \supseteq \sigma^*$ decides φ_n^0 . By (H3) the relation $\alpha \Vdash \psi^0(\bar{n})$ is Δ_1^1 , and so is $\alpha \Vdash \neg \psi^0(\bar{n})$, as $\neg \psi^0$ is ranked. The first description is then Σ_1^1 and the second Π_1^1 , because a number quantifier in front of a Δ_1^1 relation stays within Δ_1^1 ; so F is Δ_1^1 . $\square$

This is the one-bit bridge lemma of [1, Lemma 8.3] with “the computation converges with value v ” replaced by “the condition decides the sentence”. The partiality case of that lemma disappears, because deciding conditions are always dense; what was a Σ_2^0 oracle question in the Turing construction is here absorbed by genericity.

4.3 The construction

For each sentence φ of $\mathcal{L}(\omega_1^{\text{CK}}, \mathcal{G})$ let W_φ be the dense set of strings that decide φ . For each pair ψ^0, ψ^1 of ranked formulas let $E_{\psi^0, \psi^1} \subseteq \mathbb{P}$ consist of the cross-disagreements for ψ^0, ψ^1 together with the padded conditions no extension of which is a cross-disagreement. This set is dense: a condition with no cross-disagreement extension has a padded extension by Lemma 4.1(b), and that extension inherits the property.

There are countably many sets $\pi_0^{-1}[W_\varphi]$, $\pi_1^{-1}[W_\varphi]$ and E_{ψ^0, ψ^1} , and all are dense by Lemma 4.1(a). Choose an increasing sequence $p_0 \leq p_1 \leq \dots$ in $\mathbb{P}$ (that is, each extends the previous one) meeting all of them, with lengths tending to infinity, and let A and B be the unions of the first and second coordinates.

(i). Every sentence is decided by an initial segment of A , because some p_s lies in $\pi_0^{-1}[W_\varphi]$; so A is generic, and likewise B . By (H4), $\omega_1^A = \omega_1^B = \omega_1^{\text{CK}}$. Let f be a hyperarithmetical function, $q < 1$ rational and $k \in \mathbb{N}$. The set of strings of length at least $\max(k, 1)$ that disagree with f on more than the fraction q of their length is hyperarithmetical and dense, so it contains an initial segment of A by (H5). Hence $\limsup_n \rho_n(\{i : A(i) \neq f(i)\}) = 1$, and A has no hyperarithmetical coarse description. The same holds for B .

- (ii). Every initial segment of $A \triangle B$ is a subset of some $\Delta(p_s) \in \mathfrak{B}$, and $\mathfrak{B}$ is closed under subsets.
- (iii). Let $X \leq_h A$ and $X \leq_h B$. By (H4) there are ranked formulas with $X = \{n : \mathcal{M}(A) \models \psi^0(\bar{n})\} = \{n : \mathcal{M}(B) \models \psi^1(\bar{n})\}$. Some $p_s = (\alpha, \beta)$ lies in E_{ψ^0, ψ^1} . It is not a cross-disagreement: if, say, $\alpha \Vdash \varphi_n^0$ and $\beta \Vdash \neg \varphi_n^1$, then $n \in X$ and $n \notin X$ by (H2), since $\alpha \subset A$ and $\beta \subset B$ and both are generic. So $p_s = (\sigma^*, \tau^*)$ is padded and has no cross-disagreement extension. Let F be the hyperarithmetical set of Lemma 4.2. For each n some initial segment $\alpha' \supseteq \sigma^*$ of A decides φ_n^0 , and then $n \in X$ iff $\alpha' \Vdash \varphi_n^0$ iff $n \in F$. Thus $X = F$ is hyperarithmetical. The reverse inclusion in (iii) is trivial.

Finally A, B are not hyperarithmetical, being generic, so by (iii) their hyperdegrees are incomparable and have meet $\mathbf{0}_h$. This completes the proof of Theorem 1.1. $\square$

Remark 4.3 (Bounds and variants). (1) The dense sets are definable over $L_{\omega_1^{\text{CK}}}$, so the sequence (p_s) can be chosen arithmetically in $\mathcal{O}$, and then $A, B \leq_h \mathcal{O}$; relative to Z one gets $A, B \leq_h \mathcal{O}^Z$. By Theorem 1.1(i) and Spector's criterion, $\mathcal{O} \not\leq_h A$ and $\mathcal{O} \not\leq_h B$, so the witnesses lie strictly below the hyperjump. We have not checked the definability claim in detail. (2) The global-variation fusion of [1, Theorem 8.7] should give a perfect family of pairwise such sets; the forcing is then over finite frontiers, and Lemma 4.2 is unchanged. We have not written this out. (3) A two-step version should show that *every* sufficiently Cohen-generic A has such a partner B , by forcing with $\{\tau : (A \upharpoonright |\tau|, \tau) \in \mathbb{P}\}$ over A . This is not proved here either.

4.4 The corollaries

Proof of Corollary 1.2. Apply Lemma 3.2 with $Z = \emptyset$, $f = A$, $D = A$ and $E = B$; $B \in \text{CD}(A)$ because the budget forces density zero. The statement about ω_1 is Theorem 1.1(i). A representative of least Turing degree would have least hyperdegree. $\square$

Proof of Corollary 1.3. Let A, B be given by Theorem 1.1 for Z and the logarithmic budget, and put $F = J(Z) \oplus A$, $G = J(Z) \oplus B$. Then $F \approx G$, $F \equiv_{\text{T}} Z \oplus A$ and $G \equiv_{\text{T}} Z \oplus B$. By [1, Lemma 9.1], every $D \in \text{CD}(F)$ computes Z and its odd half is a coarse description of A ; so F has no coarse description in $\Delta_1^1(Z)$ by Theorem 1.1(i). Since $\Delta_1^1(F) \cap \Delta_1^1(G) = \Delta_1^1(Z)$, Lemma 3.2 applies. $\square$

5 The cone filter on the coarse degrees

For $h \in \mathbb{N}^{\mathbb{N}}$ the *coarse cones* above h are $\{f : h \leq_{\text{nc}} f\}$ and $\{f : h \leq_{\text{uc}} f\}$; the second is contained in the first.

Proof of Theorem 1.5. (a) By [1, Theorem 3.6(iv)] and the fact that $f \in \text{CD}(f)$, $f \in \mathcal{A}$ iff there are indices e, i such that Φ_e^f is a total 0–1 valued function, Φ_i^f is total and belongs to $\text{CD}(f)$, and for every $D \in \text{CD}(f)$ there is j with $\Phi_j^D = \Phi_e^f$. Membership in $\text{CD}(f)$ is arithmetical, so this is a Π_1^1 condition. Invariance under $\equiv_{\text{nc}}$ is immediate from the definition. If $f \equiv_{\text{uc}} g$ then $f \equiv_{\text{nc}} g$, so $\mathcal{A}$ is also closed under $\equiv_{\text{uc}}$; and by the spectrum identity [1, Theorem 3.1] membership in $\mathcal{A}$ is equivalent to $\text{Spec}_{\text{uc}}(f)$ having a least element.

(b) Block-code the function h itself: $J(h)(k) = h(n)$ for $2^n \leq k < 2^{n+1}$. First, $h \leq_{\text{uc}} J(h)$. From any coarse description D of $J(h)$ let $y(n)$ be the value that D takes on more than half of the n -th block if there is one, and 0 otherwise. This is a single total functional, and $y(n) = h(n)$ for all but finitely many n [1, Lemma 3.4]; a function that differs from h at finitely many places is a coarse description of h . (Coding

the *graph* of h by a set would not do here: an unbounded search in a finitely corrupted graph need not terminate, and a uniform reduction must be total on every description.)

Put $f_0 = J(h)$. Every coarse description of f_0 computes h , and $f_0 \equiv_T h$ is itself a representative, so every representative computes h and $\deg_T(h)$ is the least element of $\text{Spec}_{\text{nc}}(f_0)$; thus $f_0 \in \mathcal{A}$.

Let Z be a set with $Z \equiv_T h$, let A, B be the sets of [1, Theorem 8.7] for the oracle Z and the logarithmic budget, so that $A \approx B$, A is 1-generic relative to Z , and every set computable from both $Z \oplus A$ and $Z \oplus B$ is computable from Z . Put $f_1 = J(h) \oplus A$ and $g_1 = J(h) \oplus B$, with the block code on the even positions. Then $f_1 \approx g_1$. As in [1, Lemma 9.1], the even half of a coarse description of f_1 is a coarse description of $J(h)$ and its odd half is one of A . Suppose $g \equiv_{\text{nc}} f_1$ had least Turing degree. Then $g \leq_T f_1 \equiv_T Z \oplus A$ and $g \leq_T g_1 \equiv_T Z \oplus B$, so $g \leq_T Z$; but g computes a coarse description of f_1 , whose odd half would be a Z -computable coarse description of A , contradicting [1, Lemma 5.1]. So $f_1 \notin \mathcal{A}$. The even-half map followed by the block decoder is a uniform reduction of h to f_1 .

The consequences are immediate: each coarse cone above h contains $f_0 \in \mathcal{A}$ and $f_1 \notin \mathcal{A}$. The constructions of [1] invoked here are explicit oracle constructions and use no choice. For the last statement, under AD let $\mu_f(\mathcal{S}) = 1$ iff $\{x : [J(x)]_{\text{nc}} \in \mathcal{S}\}$ contains a Turing cone, for sets $\mathcal{S}$ of nonuniform coarse degrees. This is well defined because $x \equiv_T y$ implies $J(x) \equiv_{\text{nc}} J(y)$ [1, Proposition 3.5], and it is a countably complete ultrafilter because the Martin measure is one. It contains every nonuniform coarse cone, since $h \leq_{\text{nc}} J(h) \leq_{\text{nc}} J(x)$ whenever $x \geq_T h$. It contains $\mathcal{A}$, which is not in the cone filter. We do not claim the analogous statement for the uniform coarse degrees: it is not clear that $x \equiv_T y$ implies $J(x) \equiv_{\text{uc}} J(y)$, because a Turing reduction applied to a finitely corrupted oracle need not be total. $\square$

Remark 5.1 (Where strategy stealing fails). In Martin's proof the opponent of a winning strategy σ plays a real $x \geq_T \sigma$, and the outcome $\sigma * x$ is Turing equivalent to x . For coarse degrees the opponent would play a function g whose coarse degree is above that of a code for σ , and one would need $\sigma * g \equiv_{\text{nc}} g$. But σ 's reply at stage n depends on the *exact* initial segment of g , so a coarse description of g , which may err on a density-zero set, yields a sequence of replies that can differ from the true one at almost every stage; $\sigma * g$ is not coarsely computable from descriptions of g . Theorem 1.5 shows that this is a genuine obstruction: by C1, a coarse degree need not have a canonical representative that can be played, and the set $\mathcal{A}$ of degrees that do have one splits every cone.

Remark 5.2. The same proof shows that the set of f whose class has a least *hyperdegree* meets every coarse cone, as does its complement: use $J(h)$ and $J(h) \oplus A$, with A, B now given by Theorem 1.1 for $Z \equiv_T h$, and Lemma 3.2. By Theorem 1.4(a) this set is strictly larger than $\mathcal{A}$: it contains $\mathcal{R}(\emptyset'')$.

6 Questions

- (Q1) *A cone dichotomy for $\leq_h$.* For arbitrary X and $A \notin \text{Core}_h(X)$, is $\{D \in \mathcal{D}_X : A \leq_h D\}$ meager in the prefix-density metric? The set is Π_1^1 and so has the Baire property; if it is nonmeager it is comeager in a ball around a finite modification of any prescribed $Z \in \mathcal{D}_X$, and translating by Z turns the question into whether " $A \leq_h E \oplus Z$ for comeager many density-zero sets E in a ball" implies $A \leq_h Z$. For Cantor space this is the classical theorem that a set hyperarithmetic in comeager many reals is hyperarithmetic (Feferman; Hinman; Thomason; Kechris). We expect it to hold for the computably presented Polish space of density-zero sets, which would give exact partners and perfect families for $\leq_h$ inside *every* description class, as in [1, Theorems 4.6 and 4.12]. We have not verified the transfer.
- (Q2) *Which classes have a least hyperdegree?* By Lemma 3.1 this asks for a description hyperarithmetic in all descriptions. The block codes and the dyadic codes qualify; the generic classes of Theorem 1.1

do not. Is there a characterization by a single robust coding, as in [1, Theorem 3.6]?

- (Q3) *Jump-cones as spectra.* The dyadic code realizes $\{\mathbf{b} : \mathbf{a} \leq \mathbf{b}'\}$ as a coarse Turing spectrum. For which computable ordinals α is $\{\mathbf{b} : \mathbf{a} \leq \mathbf{b}^{(\alpha)}\}$ a coarse spectrum? The obvious double-limit coding fails, because a column whose inner limit is wrong contributes errors of positive density; we do not know whether $\alpha = 2$ is possible. A negative answer would say that coarse information is exactly “one jump robust”, and would make Core_h rather than Core the invariant that distinguishes $\mathcal{R}(A)$ from its iterates.
- (Q4) *The hyperjump.* Is there a class with no least hyperdegree all of whose representatives g satisfy $\omega_1^g > \omega_1^{\text{CK}}$, other than the relativizations $J(\mathcal{O}) \oplus A$ given by Corollary 1.3?
- (Q5) *Invariant functions.* Theorem 1.5 concerns invariant sets. Is there an analogue of Martin’s conjecture for functions on the coarse degrees that are invariant under $\equiv_{\text{nc}}$, at least on the μ_J -large set of block-code classes, where it reduces to the Turing case?

7 Status

The proofs above are complete relative to the facts (H1)–(H5) and to the results of [1] that are cited, which are themselves unrefereed. (H1)–(H5) are classical, but they were written from memory, and the exact formulation of (H3) and (H4) in the sources should be compared with the use made of them in Lemma 4.2 and in step (iii): what is needed is that forcing for the instances of a fixed ranked formula is uniformly Δ_1^1 , and that sets hyperarithmetic in a generic real are defined by ranked formulas. Remark 4.3 and question (Q1) contain unproved claims and are marked as such. Nothing here has been formalized; the Lean library `CoarseDegrees` does not contain hyperarithmetic notions. No claim of novelty is made.

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